of positive and negative particles everywhere on the membrane, that is, there is no net charge on any region of the membrane. However, when the pellet is brought near the horizon, excess negative particles move into the region below the pellet, leaving excess positive particles everywhere else on the membrane; the horizon's membrane thereby gets polarized; and the total set of field lines produced by the pellet's charges and the horizon's charges takes the form of diagram (c).

When I, as an old hand at relativity theory, heard this story, I thought it ludicrous. General relativity insists that, if one falls into a black hole, one will encounter nothing at the horizon except spacetime curvature. One will see no membrane and no charged particles. Thus, the Hanni–Ruffini description of why the pellet's electric field lines are bent can have no basis in reality. It is pure fiction. The cause of the field lines' bending, I was sure, is spacetime curvature and nothing else: The field lines bend down toward the horizon in diagram (c) solely because tidal gravity pulls on them, and not because they are being attracted to some polarized charge in the horizon. The horizon cannot possess any such polarized charge; I was sure of it. I was wrong.

Five years later Roger Blandford and a graduate student, Roman Znajek, at Cambridge University discovered that magnetic fields can extract the spin energy of a black hole and use it to power jets (the *Blandford-Znajek process*, Chapter 9 and Figure 11.4a). Blandford and Znajek also found by curved spacetime calculations that, as the energy is extracted, electric currents flow into the horizon near the hole's poles (in the form of positively charged particles falling inward), and currents flow out of the horizon near the equator (in the form of negatively charged particles falling inward). It was as though the black hole were part of an electric circuit.

The calculations showed, moreover, that the hole behaved as though it were a voltage generator in the circuit (Figure 11.4b). This black-hole voltage generator drove current out of the horizon's equator, then up magnetic field lines to a large distance from the hole, then through *plasma* (hot, electrically conducting gas) to other field lines near the hole's spin axis, then down those field lines and into the horizon. The magnetic field lines were the wires of the electric circuit, the plasma was the load that extracts power from the circuit, and the spinning hole was the power source.